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BRCAGenie: A machine learning-driven 43-gene polygenic risk score model for precision prediction of breast cancer survival

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Abstract

Background Breast cancer is one of the most prevalent malignancies globally, imposing a substantial disease burden. Its inherent heterogeneity complicates prognosis and treatment, underscoring the need for accurate survival prediction models to guide personalised clinical decision-making. Existing studies on breast cancer survival prediction models based on gene expression have largely focused on predefined subsets of genes, such as biologically-relevant, differentially-expressed or protein-coding genes, potentially overlooking critical but uncharacterised genomic contributors. To probe deeper beyond well-characterised biological pathways and differential gene expression, this study adopts an unbiased, hypothesis-free approach by conducting machine learning-based survival analysis on the entire transcriptome, rather than restricted gene subsets, to identify a novel gene signature for the development of a robust and accurate risk score model for breast cancer survival prediction.

Methods The clinical and transcriptome data from The Cancer Genome Atlas Breast Cancer (TCGA-BRCA) were analysed. Feature selection was conducted using univariate Cox, LASSO Cox regression and stepwise selection. The selected gene signature was then used to construct a risk score model, following which model performance was evaluated using AUC(t), Kaplan–Meier curves (with the Log-Rank test), Harrell's C-index and calibration plots. The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) and Gene Expression Omnibus (GSE96058) cohorts served as external validation sets.

Results Here, we present Breast Cancer Genie (BRCAGenie), a 43-gene polygenic risk score model that can reasonably predict breast cancer survival using gene expression, achieving three-year and five-year AUC(t) scores of 0.736 and 0.751 respectively in the TCGA unseen test set and significant separation of the Kaplan–Meier curves. When age at diagnosis was included, the three-year and five-year AUC(t) scores of the combined BRCAGenie model improved to 0.812 and 0.814 respectively in the unseen test set. The identified signature demonstrates promising biological relevance, with many genes supported by existing cancer literature, and others representing potentially novel biomarkers. To facilitate clinical application, the BRCAGenie calculator was developed: https://jiaweilee.shinyapps.io/breast_cancer_survival_43/.

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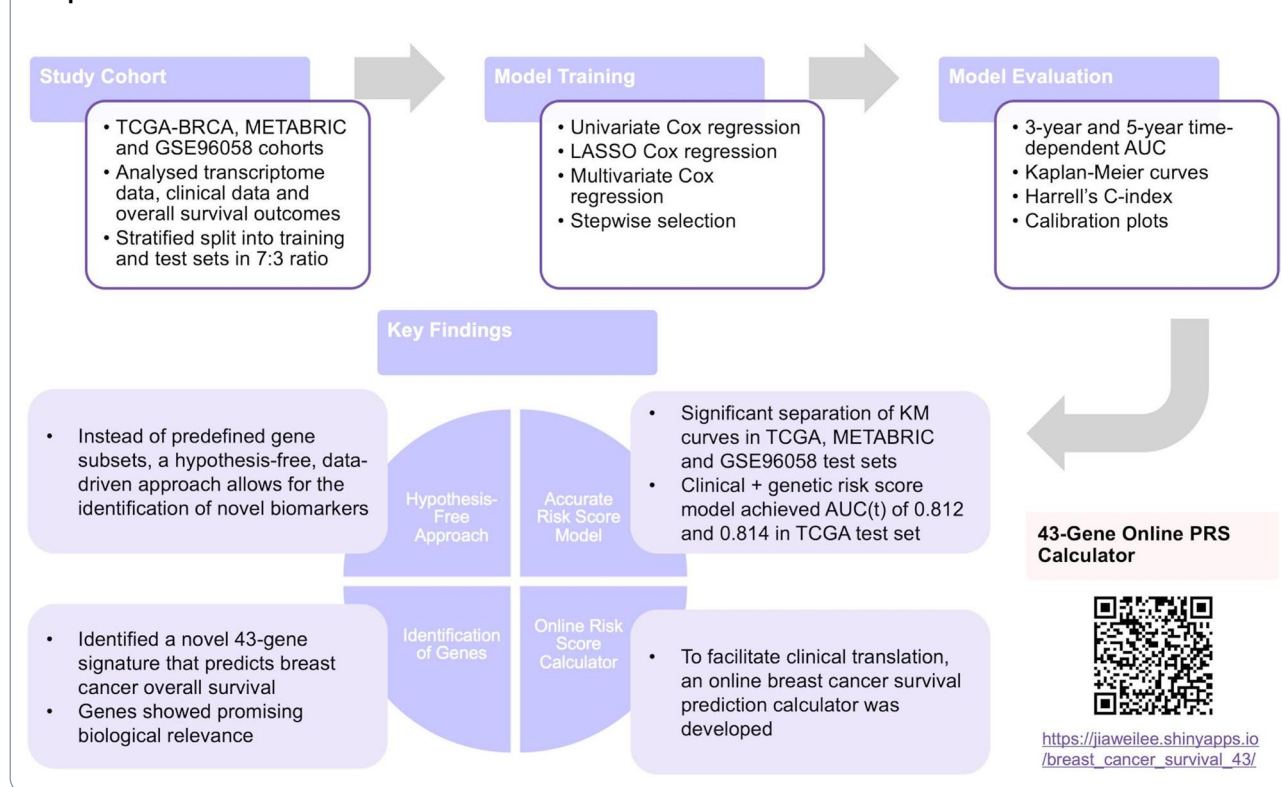


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Conclusions The development of an accurate breast cancer polygenic risk score model for survival prediction enables robust prognostic stratification for personalised clinical decision-making. Through our unbiased, hypothesis-free approach, we have identified novel breast cancer prognostic biomarkers with the potential to be clinically validated and translated into relevant diagnostics and therapeutics.

Keywords Machine learning, Biomarker, Breast cancer, Survival analysis, Polygenic risk score, Transcriptomics

Graphical abstract



Background

Breast cancer is one of the most prevalent cancers globally—as of 2020, 7.8 million women alive had a history of breast cancer [1]. In 2020, approximately 2.3 million individuals were newly diagnosed with breast cancer, with deaths amounting to 685,000 worldwide [1]. It is estimated that 13% of women will be diagnosed with breast cancer at some point during their lifetime [2]. Given its substantial disease burden, breast cancer is a pertinent issue to be targeted.

As breast cancer is a heterogeneous disease characterised by highly variable survival outcomes [3], there is a need for better tools to predict breast cancer prognosis. Predicting breast cancer survival accurately can help clinicians make personalised treatment decisions for patients, prevent excessive treatment and facilitate end-of-life planning [4, 5], optimising patients' outcomes and quality of life. Traditionally, breast cancer prognosis classification has been done mainly using clinical features or breast imaging [6–8]. However, studies have suggested that the performance of these indicators is not

satisfactory [9, 10]. Current genomic assays for breast cancer prognosis prediction include Oncotype DX and MammaPrint, yet these tools have their limitations. For instance, Oncotype DX is only suited for patients with oestrogen receptor-positive (ER+) and human epidermal growth factor receptor 2-negative (HER2-) breast cancer [11], while MammaPrint only targets early-stage breast cancer patients [12]. Moreover, given the high degree of molecular heterogeneity among breast cancer tumours, current genomic assays may not be applicable to all patients. To offer personalised and accurate recommendations for more patients, there is a need for more research into novel genomic biomarkers to improve breast cancer prognosis prediction.

In recent years, the emergence of high-dimensional omics data, especially gene expression data, holds promise in improving breast cancer prognosis prediction [13, 14]. With the advent of artificial intelligence, machine learning has emerged as a powerful tool for feature selection in high-dimensional genomics datasets, as it makes minimal assumptions and is effective even in the

presence of complex nonlinear interactions [15–17]. From the relevant genomic predictors identified through machine learning, breast cancer polygenic risk score models are often constructed such that the biomarker signatures can be clinically applicable.

While other studies have utilised machine learning on transcriptome data to construct a risk score model to predict breast cancer survival (Additional File 1, Table S1), these studies often focus on a subset of genes, such as genes related to certain biological pathways, differentially expressed genes (DEGs) or protein-coding genes [18–38]. This conventional method of analysis restricts the discovery of novel genomic biomarkers due to its underlying assumption that biomarkers predictive of cancer survival prognosis are likely to be characterised in well-known biological pathways, differentially expressed or in protein-coding regions, potentially overlooking critical but uncharacterised genomic predictors.

Hence, there is still a proportion of variance that is unexplained, prompting us to probe deeper to explore the entire genome in an unbiased manner to uncover novel breast cancer prognostic biomarkers. Hence, this study aims to utilise machine learning to identify a novel gene signature to construct a genetic risk score model that predicts breast cancer overall survival via a hypothesis-free, data-driven approach. The overall strategy of our study is summarised in Fig. 1.

Materials and methods

Study cohorts

Data from The Cancer Genome Atlas Breast Cancer (TCGA-BRCA) [39], Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) [40] and Gene Expression Omnibus (GSE96058) [41] were used in this study. The workflow for the analyses is summarised in Fig. 1.

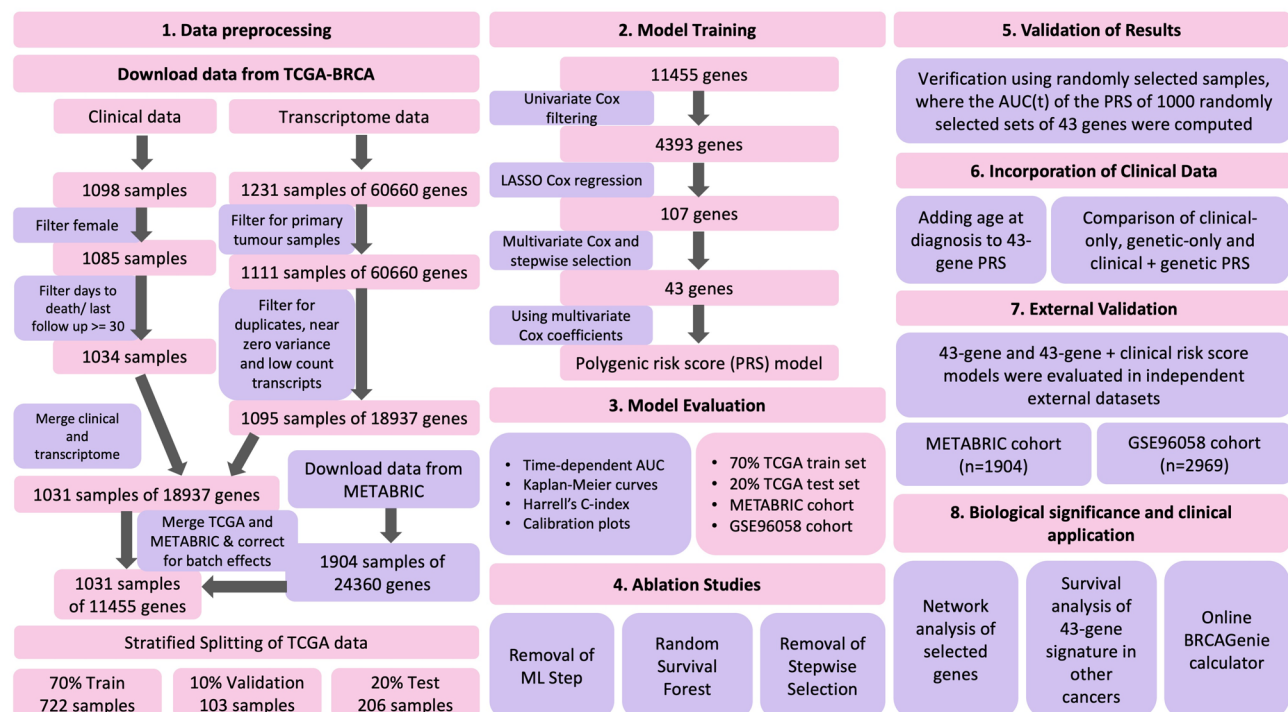


Fig. 1 Overview of methodology. The methodology consists of eight key stages: data pre-processing, model training, model evaluation, ablation studies, validation, incorporation of clinical data, external validation and biological significance and clinical application. Data pre-processing: Transcriptome and clinical data from the TCGA-BRCA cohort were retrieved using GDC query in R. After cleaning, 1031 samples with 18,937 genes remained, further filtered to 11,455 genes common to TCGA and METABRIC datasets. A stratified split (70% training, 10% validation, 20% test) was applied. Model training: Univariate survival analysis was performed on the 11,455 genes. LASSO Cox regression reduced the set to 107 genes, followed by multivariate Cox and forward stepwise selection, yielding 43 genes. The final PRS model was constructed using multivariate Cox coefficients. Model evaluation: The model was assessed using time-dependent AUC, Kaplan–Meier (KM) survival analysis, Harrell's C-index and calibration plots. Ablation studies: To justify our multi-stage feature selection pipeline, we performed ablation studies comparing a model without LASSO-Cox, a model using random survival forest for ranking of genes, a model without stepwise selection and our full proposed pipeline. Validation of results: To ensure robustness, 1000 random gene sets ($n=43$) were sampled from all features in the training set to construct PRS models for comparison. Incorporation of clinical data: Age at diagnosis was included alongside the 43 genes in the PRS model. Performance comparisons were conducted across clinical-only, genetic-only, and combined models. External validation: The METABRIC ($n=1904$) and GSE96058 ($n=2969$) cohorts served as the independent external test sets. Biological significance and clinical application: Network analysis of selected genes was performed using ConsensusPathDB and NetworkAnalyst, followed by survival analysis of the 43-gene signature in other cancers. Literature review supported the findings. Finally, an online BRCAgenie PRS calculator was developed

Data pre-processing

The relevant clinical data (1098 samples) from the TCGA-BRCA cohort was downloaded using the Genomic Data Commons (GDC) query in R. After filtering for females only and days to last follow-up or days to death more than or equal to 30 days, there were 1034 samples. Similarly, the TCGA-BRCA transcriptome data (1231 samples) was downloaded using the GDC Query. The transcripts per million (TPM)-normalized data was used. After filtering for primary tumour samples only and removing duplicates, there were 1095 samples of 60,660 genes. Subsequently, we filtered out genes that have near zero variance (based on 95% cut-off) and those with $TPM < 1$ in $\geq 70\%$ of samples. Filtering out low count transcripts (genes with $TPM < 1$ in $\geq 70\%$) was done as a quality control step to remove genes with consistently low or non-informative expression across the TCGA cohort, which are unlikely to provide reliable prognostic signals. The result was a set of 1095 samples of 18,937 genes. Merging the clinical and transcriptome data yielded 1031 samples of 18,937 genes. This set of 18,937 genes was then compared to the METABRIC dataset and further filtered to 11,455 genes which were present in both the TCGA and METABRIC datasets. The sva package was used to correct for batch effects between the two datasets (sva combat function, parametric method).

The clinical and demographic characteristics of the 1031 patients in the TCGA-BRCA cohort are summarised in Table 1. The five-year time point was chosen as the cut-off similar to previous studies on breast cancer survival analysis [42–44]. A stratified split was conducted based on the binary outcome of five-year survival to obtain the 70% training set ($n = 722$), 10% validation set ($n = 103$) and 20% test set ($n = 206$).

To ensure comparability across features, gene expression values in the TCGA training set were standardized via Z-score normalization, transforming each gene’s expression to have a mean of 0 and standard deviation of 1 based on the following formula:

$$Z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where x_{ij} is the expression of gene j in sample i , and μ_j and σ_j are the mean and standard deviation of the expression of gene j in the training set.

The same scaling parameters (μ_j and σ_j) computed from the training set were applied to transform the TCGA validation and test sets, to prevent information leakage and ensure consistent feature scaling. For the external test cohorts (METABRIC and GSE96058), gene expression data was independently standardized using its own cohort-specific means and standard deviations. These data standardization steps were essential to ensure that input features were on comparable scales for subsequent modeling.

Model training

To explore the relationship between predictor features and survival outcomes, univariate survival analysis, using the single variable Cox proportional hazards model, was conducted on the set of 11,455 genes obtained from pre-processing. Firstly, the surv_cutpoint function from the survminer R package was used to determine the optimal cutpoint for each of the 11,455 genes to categorise patients into high or low gene expression groups. Secondly, for each of the 11,455 genes, Univariate Cox proportional hazard analysis was conducted using the Surv function in the survival R package. The p value of the Log-Rank test was computed. After False Discovery Rate (FDR) correction, genes that have adjusted p -value < 0.1 were selected. This threshold was intentionally set to balance sensitivity (Type II error) and specificity (Type I error) in a high-dimensional, low-sample context. Given the large number of genes and inherent sparsity of survival signals, a less stringent FDR threshold allows for the inclusion of potentially relevant features that are further filtered through subsequent regularization and multivariate modeling steps. The subsequent multi-stage selection process effectively reduces the risk of including irrelevant genes, mitigating the risk of false positives (Type I error).

Subsequently, to reduce the number of genes to construct a parsimonious model, model training on the subset of selected genes after univariate filtering was conducted using Least Absolute Shrinkage and Selection Operator (LASSO) Cox regression, which is able to create a parsimonious model, avoiding overfitting and collinearity [45]. Using the cv.glmnet function in the glmnet R package, LASSO Cox with tenfold cross-validation was implemented, using Harrell’s C-index as the optimization metric (family = “cox”, type.measure = “C”). Genes were selected based on features that have non-zero coefficients at lambda.1se, the largest value of lambda such that error is within 1 standard error of the minimum.

Table 1 Clinical and demographic characteristics of the 1031 subjects in The Cancer Genome Atlas Breast Cancer (TCGA-BRCA) cohort

Variables	Characteristics	Overall (N = 1031)
Age at diagnosis	Mean (SD)	58.7 (12.9)
	Median [Min, Max]	58.9 [26.6, 90.0]
	Missing	14 (1.4%)
Three year survival	Survival Time ≤ 3 years	603 (58.5%)
	Survival Time > 3 years	428 (41.5%)
Five year survival	Survival Time ≤ 5 years	784 (76.0%)
	Survival Time > 5 years	247 (24.0%)

The following characteristics are summarised: Age at Diagnosis, Three Year Survival and Five Year Survival

For further feature selection and to obtain non-penalized weights for the construction of a risk score model, a standard multivariate Cox regression was applied to the genes selected from LASSO Cox. This was implemented using the `Surv` function in the `survival` R package. After ranking the genes by increasing multivariate Cox p-values, we performed forward stepwise selection on the TCGA validation set. Starting from the top-ranked gene, one gene was added at each step, and the 5-year AUC(t) was calculated. Due to high variability in AUC(t) values for small numbers of genes indicating inconsistent model performance, we focused on the region where performance began to stabilize and selected the subset corresponding to the first major peak in the AUC(t) curve, representing the most parsimonious set with peak model performance. This stepwise selection procedure avoids overfitting by excluding additional genes that do not meaningfully improve model accuracy.

Based on the final set of selected genes, the multivariate cox coefficients were used to construct the risk score model. The polygenic risk score (PRS) was calculated for each individual as the weighted sum of Z-score standardized gene expression values, using the coefficients from the multivariate Cox model as weights. Specifically:

$$PRS_i = \sum_{j=1}^p \beta_j \cdot x_{ij}$$

where PRS_i is the risk score for patient i , β_j is the multivariate Cox coefficient for gene j , x_{ij} is the Z-score standardized expression level of gene j in patient i , and p is the number of genes included in the final risk score model

Model evaluation

The same set of 43 genes, selected during the training phase, was evaluated in the TCGA training, TCGA test, METABRIC and GSE96058 datasets without re-training or re-selection. The polygenic risk score (PRS) for each sample was computed as a weighted sum of Z-score normalized gene expression values of the 43 genes in the final model, using the multivariate Cox regression coefficients from the training phase.

The time-dependent area under the curve, AUC(t), was used as an evaluation metric and computed using the `timeROC` package. The AUC(t) values and Receiver Operating Characteristic (ROC) curves were evaluated at the three-year and five-year timepoints, similar to other studies [18, 46, 47]. The Kaplan–Meier (KM) curve, one of the most established ways to analyse time-to-event data, was used for survival analysis. The Log-Rank (Mantel-Cox) test was conducted to compare the survival distributions of two independent groups. In addition,

Harrell's C-index was computed using the `survcomp` R package to assess the models' concordance between polygenic risk scores and observed survival times. The C-index values were reported with 95% confidence intervals for the training, test, and external validation sets.

Calibration plots were generated using the `pec` package. Predicted survival at 3 and 5 years were compared against observed survival frequencies to assess the calibration of the model. Calibration was performed separately for the training, test, and external validation datasets.

Ablation studies

Our multi-stage pipeline—comprising univariate Cox, LASSO-Cox regularization, followed by multivariate Cox with forward stepwise selection—was intentionally designed to balance model complexity, performance, and interpretability. To justify our multi-stage feature selection pipeline through empirical validation, we performed ablation studies comparing a model without LASSO-Cox, a model using random survival forest for ranking of genes, a model without stepwise selection and our full proposed pipeline.

Validation of results

To confirm that the observed predictive AUC(t) obtained in our PRS model was not due to random chance, 1000 independent iterations were performed, where 43 unique features were randomly sampled from all available features in the training dataset. These randomly sampled features were then used to construct PRS models, which were subsequently evaluated on both the training and unseen test datasets. The resulting AUC(t) values from each iteration were used to construct a distribution of model performance, which was then compared with our observed predictive AUC(t).

Incorporation of clinical data

Clinical data (age at diagnosis) was also incorporated as features together with the selected genes in the PRS model, and the model performance was similarly evaluated using the AUC(t), KM curves (with the Log-Rank test), Harrell's C-index and calibration plots. Another risk score model was also created using clinical data only. Subsequently, the model performance of the clinical-only, genetic-only and clinical + genetic PRS models were compared.

External validation

The METABRIC dataset was downloaded from cBioPortal and served as an independent external test set ($n = 1904$). The cohort's clinical and demographic characteristics are summarised in Table 2.

Additionally, the GSE96058 cohort was downloaded from Gene Expression Omnibus and served as another

Table 2 Clinical and demographic characteristics of the 1904 subjects in the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) cohort

Variables	Characteristics	Overall (N = 1904)
Age at diagnosis	Mean (SD)	61.1 (13.0)
	Median [Min, Max]	61.8 [21.9, 96.3]
Three year survival	Survival time ≤ 3 years	253 (13.3%)
	Survival time > 3 years	1651 (86.7%)
Five year survival	Survival time ≤ 5 years	472 (24.8%)
	Survival time > 5 years	1432 (75.2%)

The following characteristics are summarised: Age at Diagnosis, Three Year Survival and Five Year Survival

Table 3 Clinical and demographic characteristics of the 2969 subjects in the Gene Expression Omnibus (GSE96058) cohort

Variables	Characteristics	Overall (N = 2969)
Age at diagnosis	Mean (SD)	62.8 (13.1)
	Median [Min, Max]	64.0 [24.0, 96.0]
Three year survival	Survival Time ≤ 3 years	465 (15.7%)
	Survival Time > 3 years	2504 (84.3%)
Five year survival	Survival Time ≤ 5 years	1882 (63.4%)
	Survival Time > 5 years	1087 (36.6%)

The following characteristics are summarised: Age at Diagnosis, Three Year Survival and Five Year Survival

external validation set (n=2969). The characteristics of the cohort are presented in Table 3.

Clinical applicability and biological significance

A prototype of the online breast cancer PRS calculator was developed using the R shiny and plotly packages. The age at diagnosis and gene expression TPM values of the selected genes can be added as input into the calculator, and the patient’s PRS and five-year survival probability would be computed.

Over-representation analysis was conducted on the selected genes using ConsensusPathDB [48] to uncover their potential biological functions and pathways based on Gene Ontology (Gene Ontology Level 3, *p* value cut-off=0.01). To identify nodal genes and explore the functional connectivity of the gene signature, tissue-specific protein–protein interaction (PPI) network analysis was carried out using NetworkAnalyst [49]. The Ensembl gene IDs of the 43 genes were submitted via the "Gene List Input" option, selecting *H. sapiens* as the organism. The PPI network was generated using the DifferentialNet database, which captures differential protein interactions across human tissues. The PPI analysis was tissue-specific (restricted to breast mammary tissue), using the default differential interaction score filter of 15.0.

Lastly, to determine the potential applicability of BRCAGenie beyond breast cancer, survival analysis of the 43-gene signature was also conducted in other TCGA cancers, namely, liver, brain and lung cancers. We performed pan-cancer survival analysis using GEPIA2 [50].

Under “Survival Analysis”, we queried our list of 43 genes to visualize overall survival differences between high and low expression groups (stratified by median expression). Kaplan–Meier survival curves were generated for breast invasive carcinoma (BRCA), liver hepatocellular carcinoma (LIHC), brain lower grade glioma (LGG), glioblastoma multiforme (GBM), lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC).

Software used

All data analysis was conducted using R (version 4.4.1). A complete list of packages and their versions is provided in Additional File 3, Supplementary Text S1 (R Session Information). *P*<0.05 was regarded as statistically significant.

Results

Development of PRS with machine learning

Univariate Cox regression filtering (*p*<0.1) on the set of 11,455 genes yielded 4393 genes. LASSO Cox Regression was conducted on the subset of 4393 genes and based on features that have non-zero coefficients at lambda.1se, 107 genes were selected (Fig. 2a). Subsequently, a multivariate Cox regression model was constructed based on the 107 genes.

Forward stepwise selection was conducted in the validation set, and 43 genes were selected at the first major peak in the AUC(t) curve (Fig. 2b), representing the most parsimonious gene set with peak prognostic performance. Limiting the number of features avoids overfitting and increases the likelihood that the selected genes reflect key molecular drivers of disease progression, rather than noise or redundant signals. As seen in Fig. 2b, this local maximum in AUC(t) values indicates that the model achieves strong predictive performance at this point of 43 genes. After 43 genes, the AUC(t) generally declines or plateaus, suggesting that adding more genes does not improve and may even degrade model performance. Thus, the 43-gene cut-off provides an optimal trade-off between performance and simplicity.

To construct a risk score model, the PRS was calculated using the multivariate Cox coefficients of the 43 selected genes. The genes with negative multivariate Cox coefficients indicate a favourable survival prognosis, whereas those with positive coefficients are associated with a poor prognosis (Table 4).

The patients were delineated into high and low risk groups based on the median PRS score. Based on the KM curves for the training set, there is a significant difference in survival between high and low PRS groups (*p*<0.0001) (Fig. 2d, bottom). The 43-gene risk score model achieved three-year and five-year AUC(t) scores of 0.958 and 0.930 respectively as well as a C-index value of 0.921 in the training set (Fig. 2c and d). Calibration analysis

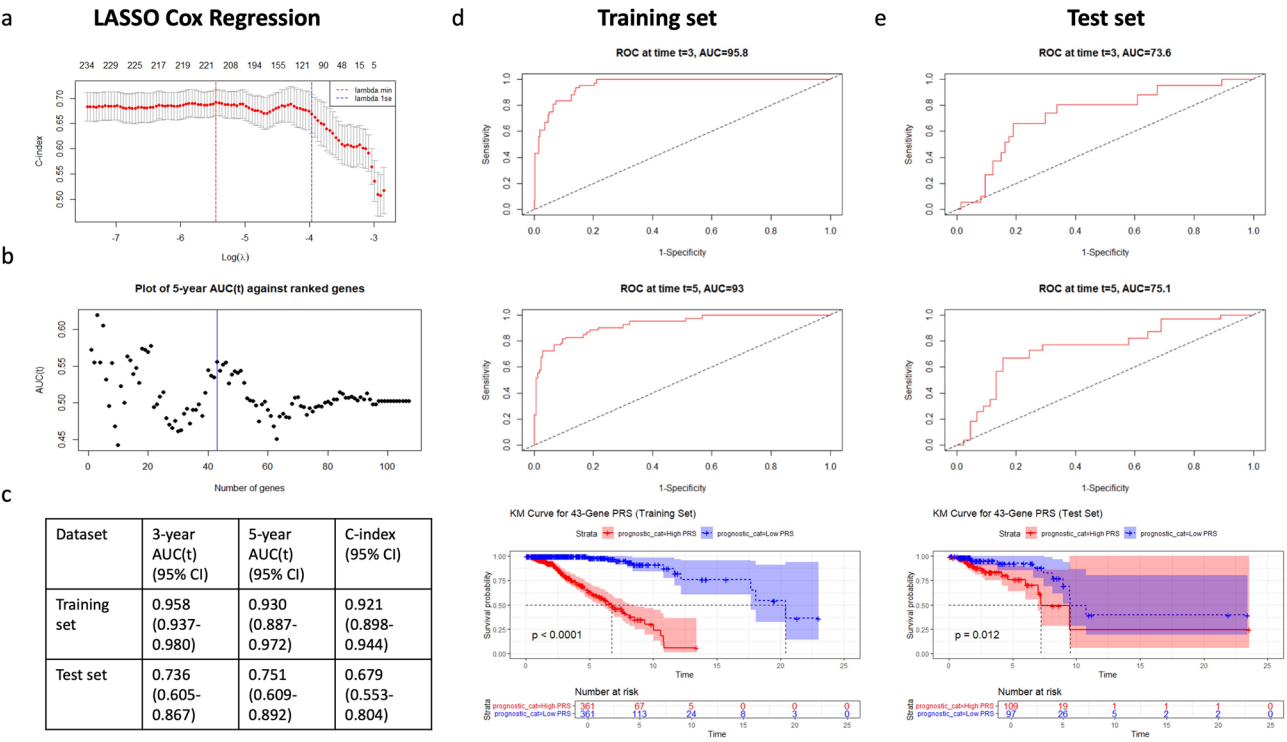


Fig. 2 Construction and performance of the 43-gene polygenic risk score (PRS) model. **a** LASSO Cox regression with tenfold cross-validation was applied to 4393 genes from univariate Cox analysis, selecting 107 genes at λ_{1se} (right dotted line). **b** Forward stepwise selection in the validation set ranked genes by multivariate Cox p -values. Five-year AUC(t) scores were plotted, with 43 genes selected at the performance peak. **c** Three- and five-year time-dependent AUC values as well as the C-index for the 43-gene model in the training and test sets. Time-dependent ROC curves and Kaplan–Meier (KM) survival analysis for the 43-gene PRS in the training **d** and test **e** sets. The Log-Rank test compared survival between high- and low-risk groups

demonstrated that the predicted survival closely matched observed survival frequencies across the range of survival groups at both 3 and 5 years in the training set (Fig. S1a). Subsequently, the PRS was calculated in the unseen test set using the same multivariate Cox coefficients and PRS cut-off value obtained in the training set. The KM curves indicate that there is a significant difference in survival between the high and low PRS groups ($p=0.012$) (Fig. 2e, bottom). A slight convergence of the survival curves is observed at around year 9; however, this does not represent a true crossover and occurs at a point where the number of patients at risk is substantially reduced, leading to greater variability in survival estimates. This slight convergence is not statistically meaningful and is likely due to censoring and reduced sample size in long-term follow-up, and does not diminish the model’s overall discriminative performance. The model maintains robust separation of high- and low-risk groups during the clinically most relevant window (0–5 years). Moreover, the three-year and five-year AUC(t) values of the 43-gene risk score model were 0.736 and 0.751 respectively in the test set, while the C-index was 0.679 (Fig. 2c and e). Moreover, calibration analysis demonstrated that apart from the lower survival groups, the predicted survival generally matched observed survival frequencies at both

3 and 5 years in the unseen test set (Fig. S1b), illustrating that the 43-gene risk score model performs well in both the training and test sets.

Ablation studies
Removal of LASSO-Cox (ML) step

As a comparison, another PRS model was also constructed via a similar methodology, but without machine learning, to assess whether machine learning enhances predictive performance, particularly in unseen test sets. Univariate cox filtering (adjusted p value cut-off=0.05) selected 69 out of 11,455 genes. Subsequently, the 69 genes were ranked based on multivariate Cox p values and forward stepwise selection was conducted. Through forward stepwise selection in the validation set, an optimum of 31 genes was selected at the performance peak (Additional File 2, Fig. S2a). The multivariate Cox coefficients were then used to construct a 31-gene polygenic risk score model (Additional File 1, Table S2). Although the results in the training set were acceptable (Additional File 2, Figs. S2b, c), the 31-gene signature showed poor model performance in the unseen test set, evident by how the three-year and five-year AUC(t) scores were 0.565 and 0.658 respectively, the C-index was 0.527, and the KM curve did not

Table 4 Final set of 43 genes selected to be included in the PRS after forward stepwise selection

Gene name	Gene symbol	Multivariate cox coefficient
Endoplasmic reticulum protein 27	ERP27	− 2.6490
NUMB like endocytic adaptor protein	NUMBL	− 2.6098
Ribosomal protein S6 Kinase B1	RPS6KB1	− 2.0842
Brain expressed X-Linked 5	BEX5	− 1.7170
Transmembrane Protein 14B	TMEM14B	− 1.6914
Solute carrier family 16 member 13	SLC16A13	− 1.5049
Cytochrome P450 family 4 subfamily F Member 11	CYP4F11	− 1.4646
Growth regulation by estrogen in breast cancer 1 like	GREB1L	− 1.2990
Apolipoprotein B mRNA Editing Enzyme Catalytic Subunit 3F	APOBEC3F	− 1.2551
Ribosomal protein S29	RPS29	− 1.1681
5'-nucleotidase domain containing 3	NT5DC3	− 1.1261
Prostaglandin E synthase 3	PTGES3	− 0.9675
Zinc finger protein 326	ZNF326	− 0.8347
Metastasis associated 1 family member 2	MTA2	− 0.8020
Syndecan 3	SDC3	− 0.6365
Xg Glycoprotein	XG	− 0.4343
Secretogranin II	SCG2	0.3120
3-Hydroxybutyrate dehydrogenase 1	BDH1	0.6032
ALX Homeobox 4	ALX4	0.6334
Cytokine Like 1	CYTL1	0.6467
ATPase H+-Transporting V1 Subunit E1	ATP6V1E1	0.6498
Surfeit 6	SURF6	0.6566
Transmembrane Protein 41A	TMEM41A	0.6600
HOX Transcript Antisense RNA	HOTAIR	0.6655
PPFIA Binding Protein 2	PPFIBP2	0.6724
Complement C6	C6	0.7013
Malignant T-Cell Amplified Sequence 1	MCTS1	0.7264
5'-Nucleotidase Ecto	NT5E	0.7425
Argininosuccinate Synthase 1	ASS1	0.7785
Transmembrane And Coiled-Coil Domains 1	TMCO1	0.7987
P21 (RAC1) Activated Kinase 6	PAK6	0.8609
Ring Finger Protein 135	RNF135	0.9085
Globoside Alpha-1,3-N-Acetylgalactos-aminyltransferase 1	GBGT1	0.9389
Transient Receptor Potential Cation Channel Subfamily M Member 2	TRPM2	1.0328
Exocyst Complex Component 1	EXOC1	1.0401
Phosphoglycerate Kinase 1	PGK1	1.1237
Cation Channel Sperm Associated 3	CATSPER3	1.1537
Peroxidasin Like	PXDNL	1.2026
Phosphofructokinase, Liver Type	PFKL	1.2867
Ectodysplasin A2 Receptor	EDA2R	1.2898
Inorganic Pyrophosphatase 2	PPA2	1.4623
Tubulin Delta 1	TUBD1	2.0995
C-C Motif Chemokine Receptor 7	CCR7	3.3420

The gene name, gene symbol and the multivariate cox coefficient of each of the 43 genes included in the PRS are shown. The genes are arranged in ascending order based on the multivariate Cox coefficient

show significant separation ($p=0.13$) (Additional File 2, Figs. S2b, d). These results highlight the crucial role of the machine learning step in selecting key genes predictive of breast cancer survival while ensuring robustness and minimising overfitting, thereby improving the predictive performance on unseen test data.

Ranking genes based on random survival forest

To compare our proposed feature selection pipeline with alternative machine learning methods, we implemented a Random Survival Forest (RSF) model using the same TCGA training and test datasets.

After univariate cox filtering and LASSO Cox regression which yielded 107 genes as in our proposed pipeline, we constructed a random survival forest model and ranked genes based on the random forest variable importance score for forward stepwise selection. Through forward stepwise selection in the validation set, an optimum of 26 genes was selected at the performance peak (Additional File 2, Fig. S3a).

The multivariate Cox coefficients were then used to construct a 26-gene polygenic risk score model (Additional File 1, Table S3). Although the results in the training set were acceptable (Additional File 2, Figs. S3b, c), the 26-gene signature showed poor model performance in the TCGA test set, evident by how the three-year and five-year AUC(t) scores were 0.556 and 0.582 respectively, the C-index was 0.526, and the KM curve did not show significant separation ($p=0.093$) (Additional File 2, Figs. S3b, d). The performance of this random survival forest model was consistently lower across multiple metrics, including the C-index and time-dependent AUC(t), compared to our proposed model, suggesting that our approach strikes a more optimal balance between interpretability, parsimony, and predictive accuracy for breast cancer survival prediction.

Removal of stepwise selection step

As a comparison, another PRS model was constructed via a similar methodology, but without stepwise selection, to assess the importance of the stepwise selection step in our feature selection pipeline. Following our proposed pipeline, univariate cox filtering selected 4393 out of 11,455 genes. Subsequently, LASSO-Cox regression was conducted, yielding a final set of 107 genes with non-zero coefficients.

Multivariate Cox modelling was conducted on the 107 genes, and the Cox regression coefficients were then used to construct a 107-gene polygenic risk score model (Additional File 1, Table S4). Although the results in the training set were acceptable (Additional File 2, Figs. S4a, b), the 107-gene signature showed poor model performance in the unseen test set, as seen from how the KM curve did not show significant separation ($p=0.62$)

(Additional File 2, Figs. S4a, c). These results highlight the crucial role of the stepwise selection step in selecting the most parsimonious gene set with peak prognostic performance, ensuring robustness and minimising overfitting by excluding additional genes that do not meaningfully improve model accuracy, thereby improving the predictive performance on unseen test data.

Verification using randomly selected samples

To verify that the performance of our 43-gene PRS model was not due to random chance, the AUC(t) values of risk score models constructed from randomly selected sets of 43 genes were computed. The results for the training set are shown in Fig. 3a and b for the three-year and five-year AUC(t) respectively, while the test set results are shown in Fig. 3c and d. For all four graphs, the AUC(t) obtained for our risk score model (red line) exceeds the 95th percentile (blue line), indicating that our PRS model performed much better than random prediction.

Incorporation of clinical data

The patient's age at diagnosis was incorporated with the 43-gene signature to construct the PRS model (Table 5). Based on the KM curves, there is a significant difference in survival between the high and low PRS groups for the training set ($p < 0.0001$) (Fig. 4a, bottom), and the clinical+genetic risk score model achieved three-year and

five-year AUC(t) values of 0.972 and 0.953 respectively as well as a C-index of 0.938 in the training set (Fig. 4a, bottom).

Subsequently, the PRS was calculated in the unseen test set using the same multivariate cox coefficients and PRS cut-off as the training set. There is a statistically significant difference in survival between the low and high PRS groups in the unseen test set based on the KM curves ($p = 0.00012$) (Fig. 4b, bottom), yielding three-year and five-year AUC(t) scores of 0.812 and 0.814 as well as a C-index of 0.741 (Fig. 4b, bottom). Calibration analysis demonstrated that the model's predicted survival generally matched the observed survival frequencies in both the training and unseen test sets (Fig. S5a, b). Thus, the clinical+genetic risk score model performs well in both the training and unseen test set.

Next, a PRS model was constructed with clinical variables only. The PRS was calculated using the Cox coefficients as outlined in Additional File 1, Table S5. The KM curves and AUC(t) scores of the clinical-only risk score model for both the training and test sets are shown in Fig. 4a and b (top). Comparing the three models, the clinical+genetic model achieved better performance (AUC(t) scores of 0.812 and 0.814 in the unseen test set) as compared to the 43-gene-only and clinical-only risk score models (Fig. 4). Overall, it is evident that

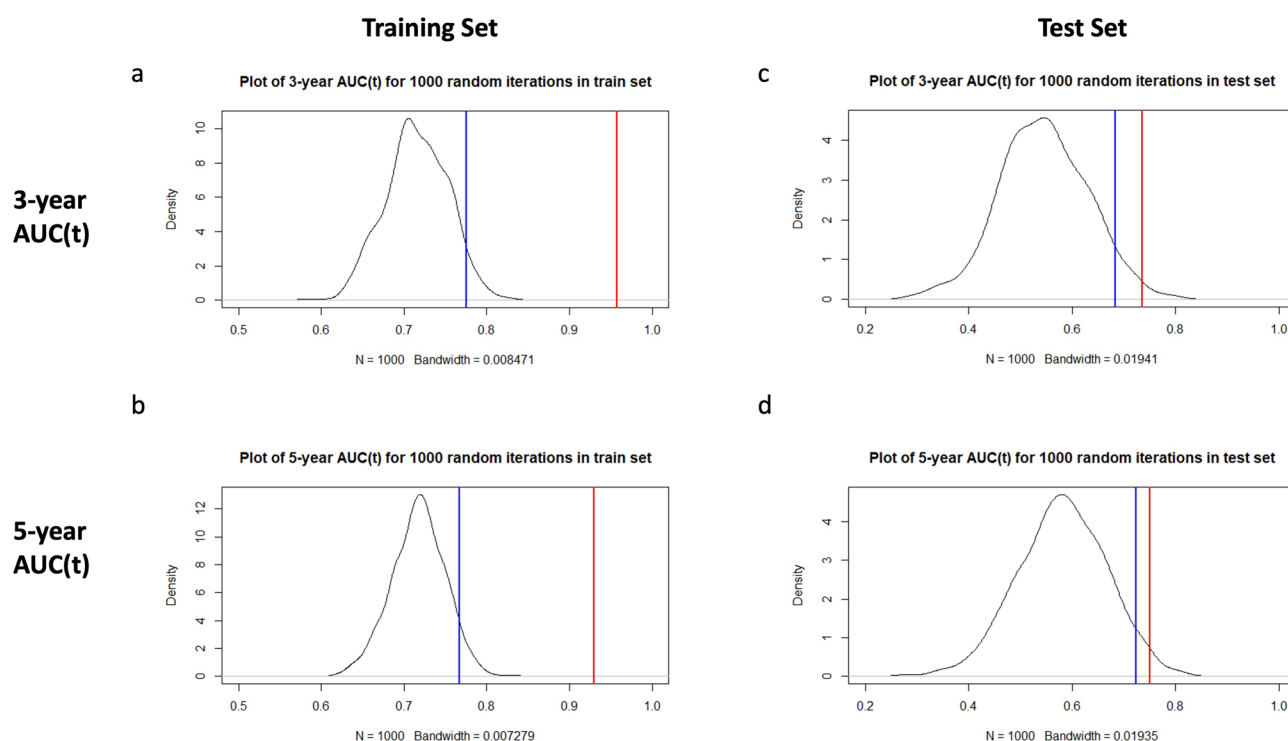


Fig. 3 The 43-gene PRS model outperformed random chance. 1000 random gene sets ($n = 43$) were sampled from all features in the training set, and PRS models were constructed. The resulting three- (a, c) and five-year (b, d) AUC(t) scores for these PRS models for the training (a, b) and test (c, d) sets were plotted. The blue vertical line shows the 95th percentile, while the red vertical line indicates the AUC(t) obtained for our risk score model

Table 5 Genetic and clinical features included in the PRS model

Variable	Multivariate coefficient	P-value
RPS6KB1	−1.0039	3.68E−03
TUBD1	0.9280	1.07E−02
PXDNL	0.5045	1.17E−07
NUMBL	−0.9295	1.55E−05
CCR7	0.4976	2.98E−03
PPA2	0.1377	4.25E−01
ALX4	0.3286	8.36E−07
PFKL	0.5750	3.32E−03
RNF135	0.4621	8.77E−05
EXOC1	0.4085	1.73E−02
APOBEC3F	−0.7288	1.16E−07
EDA2R	0.6618	1.33E−05
PGK1	0.5598	1.83E−04
CATSPER3	0.8440	2.44E−09
SLC16A13	−0.6982	3.31E−04
TRPM2	0.2822	3.62E−02
GBGT1	0.7154	5.28E−07
TMEM14B	−0.4823	2.32E−02
PTGES3	0.0636	6.82E−01
ERP27	−1.1711	1.83E−04
NT5E	0.2645	3.37E−02
BEX5	−0.6013	1.33E−03
C6	0.3617	1.37E−04
ASS1	0.0170	8.97E−01
PAK6	0.4733	3.96E−05
TMCO1	−0.0133	9.29E−01
CYTL1	0.5822	1.15E−06
PPFIBP2	0.3625	4.37E−04
NT5DC3	−0.7307	3.33E−03
MTA2	−0.3961	6.30E−03
RPS29	−0.7609	5.95E−05
GREB1L	−0.7856	2.12E−04
TMEM41A	0.5451	1.18E−04
ATP6V1E1	0.1664	2.53E−01
HOTAIR	0.3975	2.92E−04
SCG2	0.2105	4.62E−05
SURF6	0.3467	1.60E−02
ZNF326	−0.6445	3.16E−04
SDC3	−0.3120	5.36E−02
MCTS1	0.5090	2.62E−03
XG	−0.0445	6.93E−01
BDH1	0.2874	3.09E−02
CYP4F11	−0.9185	1.21E−03
Age at Diagnosis	0.0107	2.88E−01

The 43 genes and age at diagnosis were added into the multivariate Cox regression model, and the multivariate Cox coefficients were used to calculate the PRS

incorporating clinical data into the 43-gene risk score model improves model performance.

Validation on external test sets

To validate if BRCAgenie would be applicable across diverse populations, the METABRIC and GSE96058

datasets were used as external test sets. The 43-gene and 43-gene + clinical risk score models were tested in METABRIC and GSE96058 using the multivariate Cox coefficients as determined in the TCGA training set.

For the 43-gene model tested in METABRIC, the three-year and five-year AUC(t) values were 0.635 and 0.587 respectively, while the C-index was 0.552 (Fig. 5a). Based on the KM curves (Fig. 5b), there is a statistically significant difference in the survival prognosis between the low versus high PRS groups ($p = 0.00067$). When the 43-gene PRS was evaluated in GSE96058, the AUC(t) values were 0.593 and 0.590, while the C-index was 0.582 (Fig. 6a). The KM curves show a statistically significant difference in survival between the low versus high PRS groups ($p = 0.00018$) (Fig. 6b).

For the genetic + clinical model evaluated in METABRIC, the three-year and five-year AUC(t) values were 0.632 and 0.589 respectively, while the C-index was 0.557 (Fig. 5c). Based on the KM curves (Fig. 5d), there is a statistically significant difference in the survival prognosis between the low versus high PRS groups ($p = 0.00015$). For the genetic + clinical model tested in GSE96058, the three-year and five-year AUC(t) values were 0.643 and 0.631 respectively, while the C-index was 0.631 (Figs. 6c). Based on the KM curves (Fig. 6d), there is a statistically significant difference in the survival prognosis between the low versus high PRS groups ($p < 0.0001$). Hence, BRCAgenie has moderately good predictive performance when validated in the METABRIC and GSE96058 datasets. The calibration plots revealed that the 43-gene and combined PRS models showed moderate agreement in the METABRIC and GSE96058 cohorts across most risk strata (Figs. S1c and d, S5c and d). However, a consistent pattern of underestimation was observed among patients with poorer prognoses, where the model predicted lower survival probabilities than the observed. Despite this, the overall calibration remained acceptable, supporting the utility of our model.

Prototype of breast cancer PRS calculator

To facilitate clinical application, an online BRCAgenie PRS calculator was developed (https://jiaweilee.shinyapps.io/breast_cancer_survival_43/). Using age at diagnosis and the scaled TPM values of the 43 genes as input, the patient's PRS and five-year survival probability would be computed. Such information regarding the patient's predicted prognosis can facilitate data-driven and personalised clinical decision-making.

Pathway analysis

To uncover the biological significance of the 43 genes, over-representation analysis was conducted using ConsensusPathDB. The biological pathways were clustered into four main groups: metabolism and biosynthesis,

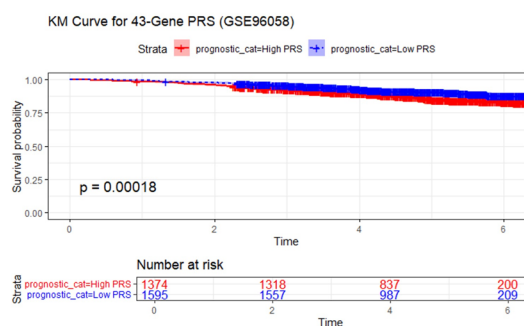
a AUC(t) and C-index for 43-Gene PRS

3-year AUC(t) (95% CI)	5-year AUC(t) (95% CI)	C-index (95% CI)
0.593 (0.546-0.639)	0.590 (0.552-0.628)	0.582 (0.548-0.616)

C AUC(t) and C-index for 43-Gene + Clinical PRS

3-year AUC(t) (95% CI)	5-year AUC(t) (95% CI)	C-index (95% CI)
0.643 (0.598-0.687)	0.631 (0.595-0.667)	0.631 (0.598-0.663)

b Kaplan-Meier Curve for 43-Gene PRS



d Kaplan-Meier Curve for 43-Gene + Clinical PRS

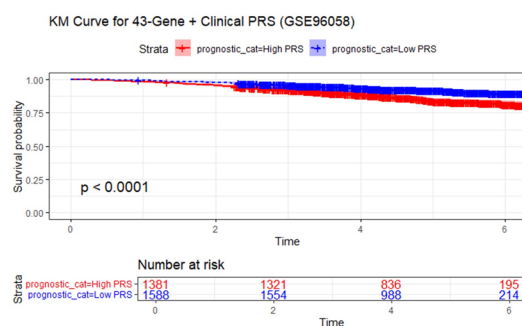
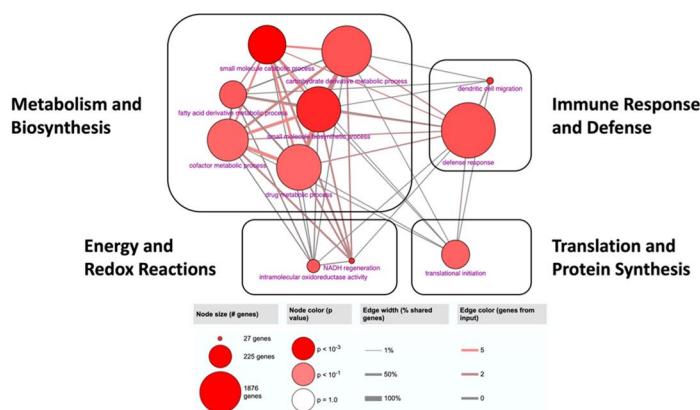


Fig. 6 External validation in the GSE96058 dataset showed moderately good performance of BRCAgenie. **a** Three- and five-year time-dependent AUC scores as well as the C-index of the 43-Gene PRS in the GSE96058 dataset. **b** KM survival curves comparing high- and low-risk groups based on the 43-Gene PRS. **c** Three- and five-year time-dependent AUC scores as well as the C-index of the 43-Gene+Clinical PRS in the GSE96058 dataset. **d** KM survival curves comparing high- and low-risk groups based on the 43-Gene+Clinical PRS

a



b

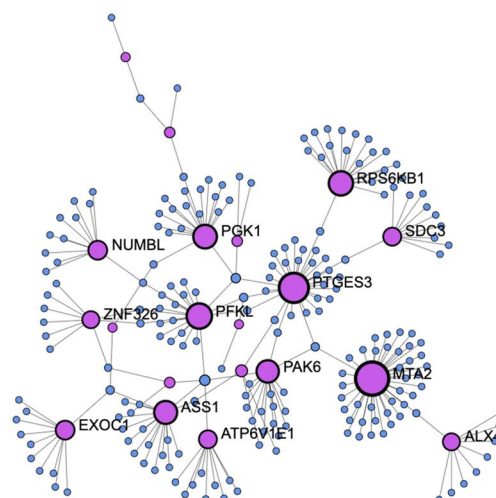


Fig. 7 Network analysis of the 43-gene signature. **a** Over-representation analysis (ConsensusPathDB) identified significant biological pathways (Gene Ontology Level 3, $p < 0.01$). Pathways were clustered into metabolism and biosynthesis, immune response and defense, energy and redox reactions and translation and protein synthesis. **b** Breast-specific protein–protein interaction network analysis (NetworkAnalyst) revealed that the genes within the signature displayed strong functional connectivity, with 13 genes (MTA2, PTGES3, PFKL, RPS6KB1, PGK1, ASS1, PAK6, EXOC1, NUMBL, ATP6V1E1, SDC3, ALX4 and ZNF326) identified as the major nodes. The purple nodes indicate genes within the 43-gene signature, while the blue nodes indicate other interacting genes/ proteins. The node size shows the degree centrality, with hub genes having a larger node size

immune response and defense, energy and redox reactions and translation and protein synthesis (Fig. 7a). Breast-specific protein–protein interaction network analysis was also carried out using NetworkAnalyst. The

network analysis revealed that the 43-gene signature showed strong functional connectivity, as 20 out of 43 genes in our signature are connected within the network (Additional File 1, Table S6). Of these, 13 genes (MTA2,

PTGES3, PFKL, RPS6KB1, PGK1, ASS1, PAK6, EXOC1, NUMBL, ATP6V1E1, SDC3, ALX4 and ZNF326) were identified as major nodes and are visualized as large purple nodes labeled with gene names in the network diagram, while the additional 7 genes appeared as smaller purple nodes, representing connected but less central genes (Fig. 7b). The remaining 23 genes in our signature are not connected in the network. These genes likely lack currently known or predicted protein–protein interactions in breast tissue at this specificity level. Nevertheless, these genes represent novel and underexplored biomarkers that merit further study in the breast cancer context.

Survival analysis in other cancers

To explore the potential applicability of BRCAGenie beyond breast cancer, survival analysis of the 43-gene signature was also conducted in other TCGA cancers. The cancers analysed include breast invasive carcinoma (BRCA), liver hepatocellular carcinoma (LIHC), brain lower grade glioma (LGG), glioblastoma multiforme (GBM), lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC), as the liver, brain and lung are common sites of metastasis for breast cancer [51, 52]. Apart from BRCA, the KM curves showed significant survival differences between the high and low groups of the 43-gene signature (log rank $p < 0.05$) for LIHC, LGG

and GBM (Fig. 8), suggesting that the PRS could potentially be applied to brain and liver cancers as well.

However, our 43-gene signature did not show significant survival differences for LUAD and LUSC. We hypothesize that this may reflect the distinct molecular landscape of lung cancers. LUAD is often driven by mutations in genes such as EGFR or KRAS [53–55], while LUSC is primarily characterized by alteration in TP53 [56–58]. Notably, these key driver genes are not included in our 43-gene signature, which may explain the limited performance observed in these cancers. These findings suggest that while our polygenic risk score may capture pan-cancer survival signals, it may not generalize to all cancer types, reinforcing the need for cancer-type-specific risk models.

Discussion

In this study, we have developed BRCAGenie, a polygenic risk score (PRS) model trained on TCGA data and validated in the TCGA, METABRIC and GSE96058 cohorts. BRCAGenie demonstrates strong predictive performance for breast cancer survival. While there is an attenuation in predictive performance when testing the model in the external test cohorts, as compared to in the training and internal test sets, this could be due to differences in patient demographics and clinical compositions. Despite

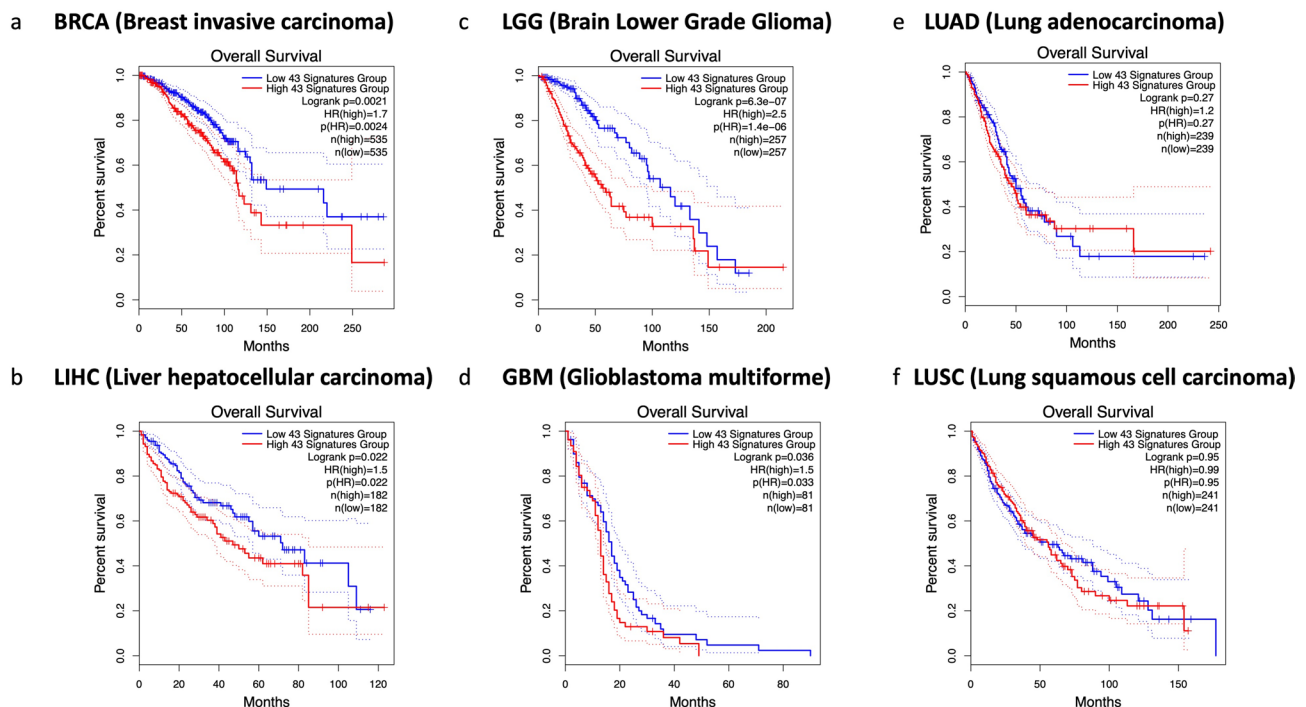


Fig. 8 Kaplan–Meier (KM) survival plots of the 43-gene signature in various TCGA cancers. Pan-cancer survival analysis was conducted using GEPIA2 to visualize overall survival differences between high and low expression groups of our 43-gene signature for **a** breast invasive carcinoma, **b** liver hepatocellular carcinoma, **c** brain lower grade glioma, **d** glioblastoma multiforme, **e** lung adenocarcinoma and **f** lung squamous cell carcinoma. The KM curves shows significant survival differences between the high and low groups for breast, brain and liver cancers. The red curve shows the high 43 signatures group while the blue curve indicates the low 43 signatures group

Table 6 Model performance of our 43-gene risk score model compared with other published studies

Study	Gene Number	Three-year AUC(t)	Five-year AUC(t)
Construct ceRNA network and risk model of breast cancer using machine learning methods under the mechanism of Cuproptosis [19]	4	0.604	0.550
Establishment of a novel CNV-related prognostic signature predicting prognosis in patients with breast cancer [59]	9	0.630	0.580
Construction and validation of a newly prognostic signature for CRISPR-Cas9-based cancer dependency map genes in breast cancer [27]	6	0.649	0.618
Identification of a 6-gene signature for the survival prediction of breast cancer patients based on integrated multi-omics data analysis [60]	6	0.660	0.630
A novel hypoxia- and lactate metabolism-related signature to predict prognosis and immunotherapy responses for breast cancer by integrating machine learning and bioinformatic analyses [26]	5	0.671	0.638
A novel disulphidptosis-associated expression pattern in breast cancer based on machine learning [22]	4	0.687	0.644
Insights into a Machine Learning-Based Palmitoylation-Related Gene Model for Predicting the Prognosis and Treatment Response of Breast Cancer Patients [28]	11	0.743	0.644
Identification of prognostic genes for breast cancer related to systemic lupus erythematosus by integrated analysis and machine learning [32]	3	0.690	0.690
Development of a Ten-lncRNA Signature Prognostic Model for Breast Cancer Survival: A Study with the TCGA Database [61]	10	0.717	0.717
A six-long non-coding RNAs signature as a potential prognostic marker for survival prediction of ER-positive breast cancer patients [62]	6	0.725	0.717
Integration of Bioinformatics and Machine Learning to Identify CD8+T Cell-Related Prognostic Signature to Predict Clinical Outcomes and Treatment Response in Breast Cancer Patients [23]	2	0.754	0.727
Development and validation of a prognostic model and gene co-expression networks for breast carcinoma based on scRNA-seq and bulk-seq data [34]	10	0.772	0.736
43-gene risk score model in our study	43	0.736	0.751

The three-year and five-year AUC(t) of our 43-gene risk score model in the test set was compared with other studies that used TCGA-BRCA for the test set and reported the three-year and five-year AUC(t) scores. The rows are sorted based on five-year AUC(t) scores

these differences between cohorts, our model still retains useful predictive power beyond the training data. In particular, Kaplan–Meier survival analysis showed that BRCAGenie consistently stratified patients into distinct prognostic groups in the TCGA unseen, METABRIC and GSE96058 cohorts (log-rank $p < 0.05$), supporting our model's prognostic relevance in independent datasets. The robustness of our model is strengthened by its validation against randomly selected 43-gene subsets, confirming that its predictive power is not due to random chance. Moreover, we compared the performance of BRCAGenie to other genetic risk score models predicting breast cancer overall survival, utilizing TCGA-BRCA for the test set and reporting three-year and five-year AUC(t) scores (Table 6). Notably, our 43-gene PRS model achieved the highest five-year AUC(t) and ranked third for three-year AUC(t) among the 13 models assessed. Given the clinical importance of five-year survival as a prognostic indicator, the superior performance of BRCAGenie in this metric highlights its potential clinical relevance.

Nevertheless, when comparing our model performance to prior studies, it is important to consider differences in the size and composition of test datasets. For example, some works used a 60/40 training/test split, whereas we adopted a 70/10/20 training/validation/test split to allow for a separate validation set for model tuning. A 60/40 ratio could allow for more rigorous testing and evaluation of model generalization [63], while our 70/10/20 design allocates more data to training and prioritizes robust validation at the cost of a smaller test set. These differences in dataset partitioning should be taken into account when comparing model performance.

Further improvements in model performance were observed when clinical data was integrated with our 43-gene PRS, yielding three-year and five-year AUC(t) values of 0.812 and 0.814 in the unseen test set. This aligns with existing research demonstrating that the incorporation of clinical variables enhances prognostic predictions across various malignancies [14, 37]. Integrating gene expression with clinical data is crucial for capturing the intricate biological mechanisms underpinning cancer progression [37].

In addition to its predictive performance, the value of BRCAGenie lies in its clinical interpretability and computational efficiency. It is implemented as a polygenic risk score (PRS) in which the contribution of each gene is defined by its Cox regression coefficient. This transparent, additive structure enhances interpretability, allowing clinicians to understand and explain how specific genes drive a patient's overall predicted risk. Furthermore, our model is computationally efficient as once the coefficients are fixed, the PRS is computed by a simple weighted sum of gene expressions, which is extremely fast and memory-light, allowing instantaneous risk calculation for each

patient. Compared to more complex approaches such as random survival forests or deep learning, our approach is lightweight, scalable, and readily deployable in clinical settings, especially resource-limited environments.

Our study adopted a hypothesis-free, data-driven approach instead of focusing on predefined subsets of genes, such as differentially-expressed or pathway-specific genes. By employing machine learning-based survival analysis across the entire transcriptome, we identified novel prognostic biomarkers that might have been overlooked in conventional analyses. Traditional methodologies assume that only highly differentially expressed genes or those linked to known biological pathways would contribute to breast cancer survival prediction. In contrast, our whole-genome approach uncovered novel candidate biomarkers with prognostic potential.

Comparisons with established breast cancer genomic assays, including Oncotype DX, MammaPrint, and EndoPredict [64, 65], revealed no overlap between the 43 genes in BRCAGenie and those used in these assays. This discrepancy likely arises because traditional assays prioritize a small subset of genes that are directly involved in proliferation, hormone receptor status and known breast cancer pathways, whereas BRCAGenie was developed using a hypothesis-free, machine learning-driven approach that captures a broader range of genomic signals. Our model identified genes that may not be conventionally associated with breast cancer but could play significant roles in survival outcomes, potentially representing underexplored prognostic biomarkers.

Furthermore, among the previously reported machine learning-based breast cancer survival models leveraging transcriptome data (Additional File 1, Table S1), only two genes from our 43-gene signature, Transmembrane Protein 14B and Exocyst Complex Component 1, have been included in prior risk score models [27, 34]. The limited overlap with existing models suggests that our hypothesis-free approach successfully identified novel breast cancer prognostic biomarkers, providing a strong foundation for further validation and future translational applications.

The biological relevance of the identified 43-gene signature is underscored by pathway analysis, which highlights that these genes are predominantly associated with metabolism and biosynthesis, immune response and defense, energy and redox reactions and translation and protein synthesis (Fig. 7a). Dysregulated metabolism, immune evasion and aberrant protein synthesis are key hallmarks of cancer [66–68], while redox regulation interacts with the tumour microenvironment to drive cancer progression [69]. The strong association of these pathways to breast cancer reinforces the potential functional significance of the identified genes. Moreover, the functional connectivity of the 43-gene signature is

evident from the protein–protein interaction network analysis (Fig. 7b), strengthening the potential of this gene set as a robust prognostic signature for breast cancer. The protein–protein interaction network also identified 13 hub genes within the BRCAGenie signature, and these genes in particular should be investigated further with regards to their roles in breast cancer.

A comprehensive literature review was also conducted to examine the relevance of these 43 genes in breast and other cancers (Table 7). Each gene was classified as indicative of good or poor prognosis based on its coefficient in BRCAGenie, and this classification was compared with literature findings. Approximately 70% of the genes (30 out of 43) in the signature are supported by prior cancer literature, including breast cancer-specific studies, substantiating the biological validity of our model. We acknowledge that one gene—Xg Glycoprotein—has not been previously reported in association with cancer. This gene may be a novel breast cancer biomarker, warranting further experimental investigation. To facilitate clinical translation, we have developed an online BRCAGenie PRS calculator that allows clinicians and researchers to predict breast cancer survival prognosis using our 43-gene model. However, the current implementation requires users to input scaled Transcripts Per Million (TPM) values, which may not be practical as it necessitates next-generation sequencing followed by data normalization. Future improvements to the PRS calculator will incorporate alternative quantification methods, such as RT-qPCR, to enhance its usability and accessibility in diverse clinical and research settings.

Despite the promising results, our study has limitations. The TCGA-BRCA training cohort is predominantly composed of individuals of European descent, which may limit the generalisability of our model to diverse populations. We did not perform ancestry-stratified analyses because the sample size for other racial groups were limited, resulting in reduced statistical power and unstable survival estimates. Additionally, the METABRIC cohort lacks corresponding race information, further limiting cross-cohort ancestry-specific comparisons. We acknowledge that the underrepresentation of diverse populations remains a challenge in genomic research and may impact the generalizability of prognostic models. Moving forward, future research should focus on validating the 43-gene PRS in larger, more ancestrally diverse and multi-platform cohorts to ensure its broad applicability.

While we validated our 43-gene prognostic signature in three public datasets (TCGA, METABRIC and GSE96058), we did not include an independent institutional cohort. As such, the generalizability of our PRS remains to be demonstrated. Nevertheless, we emphasize that our findings provide a necessary foundation for

Table 7 Summary of literature review of the 43 genes identified in our study

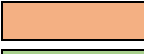
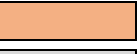
Gene name	Gene Symbol	PRS Coefficient	Good/poor prognostic marker (PRS)	Good/poor prognostic marker (Lit search)	References
C–C Motif Chemokine Receptor 7	CCR7	3.3420			[70–72]
Tubulin Delta 1	TUBD1	2.0995			[73, 74]
Inorganic Pyrophosphatase 2	PPA2	1.4623			[75, 76]
Ectodysplasin A2 Receptor	EDA2R	1.2898			[77, 78]
Phosphofructokinase, Liver Type	PFKL	1.2867			[79, 80]
Peroxidasin Like	PXDNL	1.2026			[81, 82]
Cation Channel Sperm Associated 3	CATSPER3	1.1537			[83]
Phosphoglycerate Kinase 1	PGK1	1.1237			[84–88]
Exocyst Complex Component 1	EXOC1	1.0401			[27, 89]
Transient Receptor Potential Cation Channel Subfamily M Member 2	TRPM2	1.0328			[90–92]
Globoside Alpha-1,3-N-Acetylgalactosaminyltransferase 1	GBGT1	0.9389			[93]
Ring Finger Protein 135	RNF135	0.9085			[94, 95]
P21 (RAC1) Activated Kinase 6	PAK6	0.8609			[96, 97]
Transmembrane And Coiled-Coil Domains 1	TMCO1	0.7987			[98, 99]
Argininosuccinate Synthase 1	ASS1	0.7785			[100–103]
5'-Nucleotidase Ecto	NT5E	0.7425			[104–107]
Malignant T-Cell Amplified Sequence 1	MCTS1	0.7264			[108–110]
Complement C6	C6	0.7013			[111]
PPFIA Binding Protein 2	PPFIBP2	0.6724			[112–114]
HOX Transcript Antisense RNA	HOTAIR	0.6655			[115–119]
Transmembrane Protein 41A	TMEM41A	0.6600			[120]
Surfeit 6	SURF6	0.6566			[121]
ATPase H+Transporting V1 Subunit E1	ATP6V1E1	0.6498			[122]
Cytokine Like 1	CYTL1	0.6467			[123–125]
ALX Homeobox 4	ALX4	0.6334			[126–130]
3-Hydroxybutyrate Dehydrogenase 1	BDH1	0.6032			[131, 132]
Secretogranin II	SCG2	0.3120			[133–136]
Xg Glycoprotein	XG	−0.4343			

Table 7 (continued)

Gene name	Gene Symbol	PRS Coefficient	Good/poor prognostic marker (PRS)	Good/poor prognostic marker (Lit search)	References
Syndecan 3	SDC3	−0.6365			[137]
Metastasis Associated 1 Family Member 2	MTA2	−0.8020			[138–141]
Zinc Finger Protein 326	ZNF326	−0.8347			[142–146]
Prostaglandin E Synthase 3	PTGES3	−0.9675			[147–150]
5'-Nucleotidase Domain Containing 3	NT5DC3	−1.1261			[151, 152]
Ribosomal Protein S29	RPS29	−1.1681			[153, 154]
Apolipoprotein B mRNA Editing Enzyme Catalytic Subunit 3F	APOBEC3F	−1.2551			[155, 156]
Growth Regulation By Estrogen In Breast Cancer 1 Like	GREB1L	−1.2990			[157]
Cytochrome P450 Family 4 Subfamily F Member 11	CYP4F11	−1.4646			[158–160]
Solute Carrier Family 16 Member 13	SLC16A13	−1.5049			[161, 162]
Transmembrane Protein 14B	TMEM14B	−1.6914			[34, 163]
Brain Expressed X-Linked 5	BEX5	−1.7170			[164]
Ribosomal Protein S6 Kinase B1	RPS6KB1	−2.0842			[165–167]
NUMB Like Endocytic Adaptor Protein	NUMBL	−2.6098			[168]
Endoplasmic Reticulum Protein 27	ERP27	−2.6490			[169–172]

For the PRS column, each gene was colour coded based on its coefficient in the 43-gene PRS. If the coefficient is positive, it means that the gene is a marker of poor prognosis (indicated in orange); if the coefficient is negative, it means that the gene is a marker of good prognosis (indicated in green). For the literature search column, each gene was colour coded based on its reported associations with cancer (especially breast cancer). If the gene has been shown to be overexpressed in cancers or found to be indicative of poor prognosis, it is coded as orange; if the gene has been shown to be under expressed in cancers or found to be protective against cancers, it is coded as green. Light grey indicates that currently, there has been no studies found regarding this gene and its association with cancer. Dark grey indicates that there is contradictory evidence regarding whether this gene is a good or poor prognostic marker for cancer. The relevant papers are cited. The genes are arranged in descending order based on the PRS coefficient

future efforts, such as the validation of our polygenic risk score in institutional cohorts.

Furthermore, while calibration analysis generally displayed good agreement between predicted and observed survival probabilities, a consistent pattern of underestimation was observed among the low survival groups. This discrepancy is potentially due to limited representation of high-risk individuals in the training cohort. Hence, this underscores the need for model refinement and training with larger cohorts to improve calibration and model performance.

Additionally, we acknowledge the limitation of using only in-silico methods, and note that future work should prioritize experimental interrogation, such as siRNA or CRISPR-mediated knockdown in breast cancer cell lines, to assess the functional significance of these identified

genes. Furthermore, approaches such as pathway activity inference may provide complementary insights into pathway-level mechanisms. BRCAGenie could also be explored for prognostic applications beyond breast cancer, particularly in malignancies such as brain and liver cancers, where its potential utility has already been suggested.

In the future, this biomarker signature could be translated into a clinically deployable prognostic tool, such as a lab-on-a-chip device, enabling rapid and accurate breast cancer prognosis. Such advancements could enhance personalised treatment strategies, improve patient outcomes, and refine risk stratification in clinical oncology.

Conclusions

In summary, BRCAGenie represents a robust, machine learning-driven polygenic risk score model that demonstrates strong predictive performance for breast cancer survival, outperforming several existing genetic risk models. The integration of clinical data further enhances predictive accuracy, highlighting the translational relevance of combining molecular and clinical data. By leveraging a hypothesis-free, data-driven approach, we have identified novel prognostic biomarkers beyond traditional genomic assays, underscoring the potential of whole-transcriptome analyses in precision oncology.

While our findings offer promising avenues for personalized breast cancer prognosis, further validation in diverse cohorts and experimental studies will be essential for clinical deployment. Future advancements, including the adaptation of our model into accessible prognostic tools, could significantly refine risk stratification and improve patient outcomes in oncology.

Abbreviations

BRCAGenie	Breast Cancer Genie
TCGA-BRCA	The Cancer Genome Atlas Breast Cancer
METABRIC	Molecular Taxonomy of Breast Cancer International Consortium
GDC	Genomic Data Commons
LASSO	Least Absolute Shrinkage and Selection Operator
AUC(t)	Time-dependent Area Under the Curve
AJCC	American Joint Committee on Cancer
DEGs	Differentially Expressed Genes
TPM	Transcripts Per Million
FDR	False Discovery Rate
PRS	Polygenic Risk Score
ROC	Receiver Operating Characteristic
KM	Kaplan–Meier
GEPIA2	Gene Expression Profiling Interactive Analysis 2
C-index	Concordance Index

Supplementary Information

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Additional file 1.

Additional file 2.

Additional file 3.

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Author contributions

Conceptualisation: CGL and SSC; methodology: CGL, JWL, AJWL, CW, JHN, LJJ; formal analysis: JWL; writing—original draft preparation: JWL; writing—review and editing: CGL; supervision: CGL; project administration: CGL; All authors have read and agreed to the final manuscript.

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Data availability

The data and code supporting the findings of this study are publicly available via: <https://github.com/LCFGChipmunt/brcagenie>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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